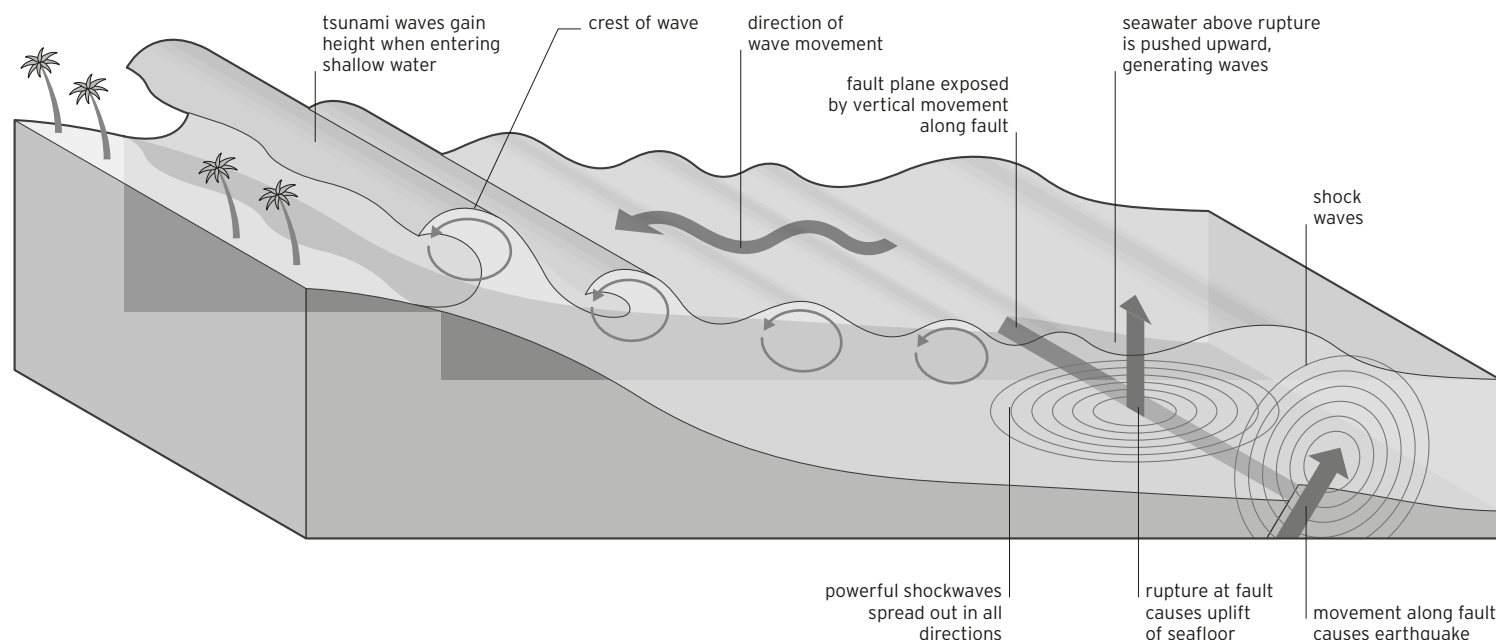


TSUNAMI

Sometimes mistakenly called a tidal wave, a tsunami is not created by tidal action. It is a fast-moving ocean wave generated by an earthquake, volcanic eruption, or landslide above or below water. If a tsunami moves into shallow

water it gains height rapidly, in extreme cases reaching a height of tens of feet and causing coastal destruction on a vast scale. The 2004 Indian Ocean tsunami killed at least 230,000 people in 14 countries.



TROPOPAUSE The boundary between the troposphere and the stratosphere, at altitudes of about 10 miles (16km) at the equator and 5 miles (8km) over the poles. Above the tropopause, the air gets warmer with height. See also *stratosphere*, *troposphere*.

TROPOSPHERE The lowest, most oxygen-rich, layer of the atmosphere, where most weather events take place. The upper limit of the troposphere is lower at the poles than at the equator. See also *tropopause*.

TROUGH In meteorology, a long, relatively narrow area of low atmospheric pressure. See also *front*.

TSUNAMI See panel, above.

TUFF An igneous rock made of fine-grained pyroclastic material, such as ash, produced by a violent volcanic eruption. See also *pyroclastic deposit*.

TUNDRA The treeless biome of low-growing, cold-tolerant

plants lying north of the taiga in northern Europe, Asia, and North America. See also *biome*, *taiga*.

TURBIDITE The sediment deposited by a turbidity current.

TURBIDITY CURRENT A rapidly moving flow of sediment-laden water, such as a flow down a continental slope onto the ocean floor.

TYPHOON See *hurricane*.

U

ULTRAVIOLET Invisible radiation with a short wavelength. Ultraviolet is a major component of sunlight. See also *infrared*.

UNDERSTORY The layer of smaller trees and shrubs growing mainly in the shade beneath a forest canopy. See also *canopy*.

UPWELLING The process whereby cold, and usually nutrient-rich, water from ocean depths rises toward the surface.

V

VALLEY FOG A type of fog that can form in low-lying areas, on cold, still nights. Heavy, chilled air settles in such places, and the water vapor it contains condenses when the temperature falls below the dew point. See also *radiation fog*.

VALLEY GLACIER A stream of flowing ice that is confined within the sides of a valley, often a V-shaped preglacial valley that is modified by the erosive action of the ice. See also *glacier*.

VEIN A fracture in rock containing mineral deposits. Veins are usually formed when minerals are

precipitated from a hot liquid in which they are dissolved. See also *hydrothermal vein*.

VENTIFACT A desert rock or pebble that has been polished by the action of windblown sand.

VERTICAL TRANSPORT The ascent (upwelling) or descent of nutrient-rich water in the oceans.

VOLATILE A term used to describe water, carbon dioxide, and some other gases dissolved in magma that form bubbles at low pressure near the surface. When the bubbles combine, they can make an eruption more explosive, showering sprays of lava. See also *eruption* (panel).

VOLCANIC ARC A chain of volcanoes formed above a subduction zone and arranged in an arc shape. The volcanoes may form a chain of islands in the ocean. See also *island arc*, *subduction*, *tectonic plate*.